

Unit 16

HYDROXY COMPOUNDS

43 items · 87 marks

Syllabus sections in this unit

16.1 Alcohols

Unit 16: Hydroxy compounds

1. M/J 26/P21/Q4(a)(i)

Context: part (a) (introduction)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

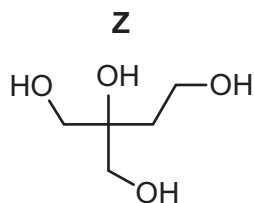


Figure 4.1

- (i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

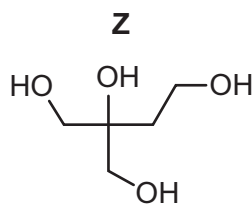
colour of mixture after the reaction

[1]

2. M/J 26/P21/Q4(b)(i)

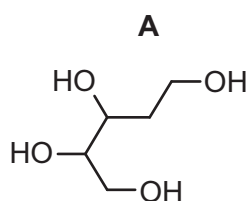
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

**Figure 4.2**

- (i)** Complete Table 4.1 to show what is observed when separate samples of **A** are added to each reagent. State 'none' if no reaction occurs.

Table 4.1

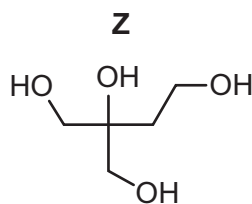
reagent	observation with A
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

3. M/J 26/P21/Q4(b)(iii)

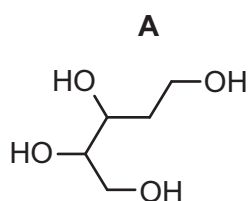
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

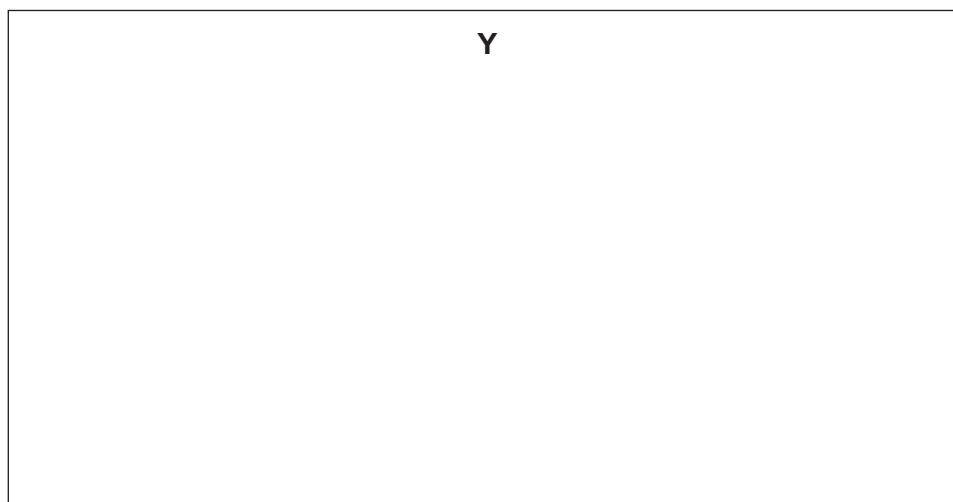
Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

**Figure 4.2**

- (iii)** Compound **A** is heated under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ to produce organic compound **Y**.

Draw the structure of **Y**.



[2]

4. M/J 26/P22/Q4(a)

4 (a) Figure 4.1 shows two reactions of compound **D**.

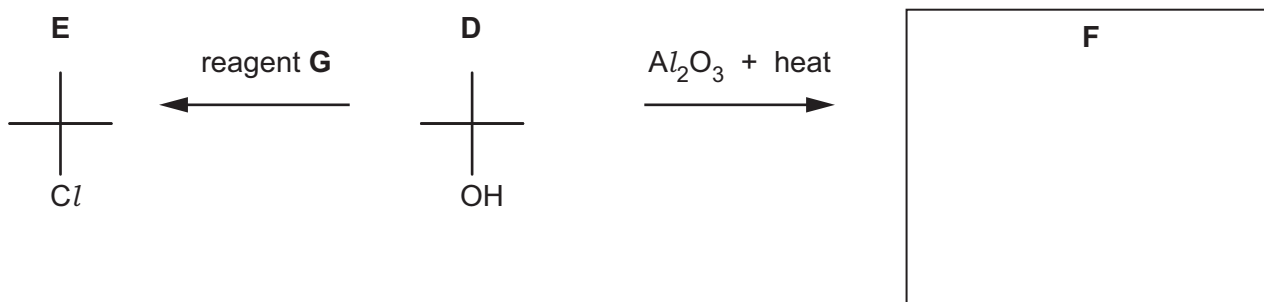


Figure 4.1

(i) **D** reacts with a single reagent, **G**, to produce compound **E**. Identify reagent **G**.

[1]

(ii) **D** is heated with Al_2O_3 to produce organic compound **F**. Identify the role of Al_2O_3 in this reaction.

[1]

(iii) Draw the structure of **F** in Figure 4.1.

[1]

5. M/J 26/P22/Q4(b)(i)

Context: part (b) (introduction)

(b) The structure of compound **X** is shown in Figure 4.2.

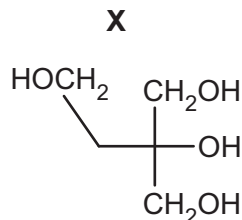


Figure 4.2

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

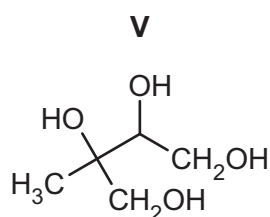
colour of mixture after the reaction

[1]

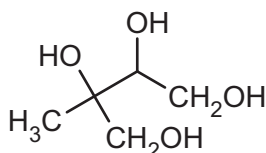
6. M/J 26/P23/Q4(a)(i)-(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

- (a) (i) Complete Table 4.1.

Table 4.1

	primary hydroxyl groups	secondary hydroxyl groups	tertiary hydroxyl groups
number of groups in one molecule of V			

[2]

- (ii) Complete Table 4.2 to show what is observed when separate samples of **V** are added to each reagent. State 'none' if no reaction occurs.

Table 4.2

reagent	observation with V
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

7. M/J 26/P23/Q4(b)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

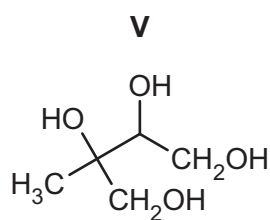


Figure 4.1

- (b)** Figure 4.1 is repeated for use in **4(b)**.

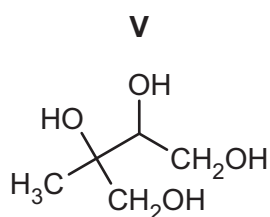


Figure 4.1

V is completely oxidised by heating under reflux with an excess of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce organic compound **Z**.

- (i)** Describe the colour change in this reaction.

colour of mixture at start of the reaction

colour of mixture after the reaction

[1]

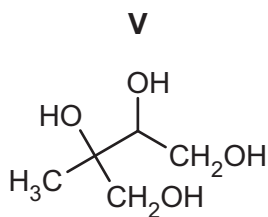
- (ii)** Draw the structure of **Z**.

[1]

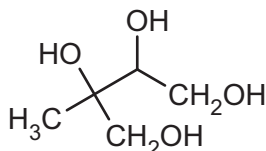
8. M/J 26/P23/Q4(c)(iii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

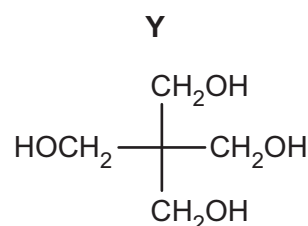
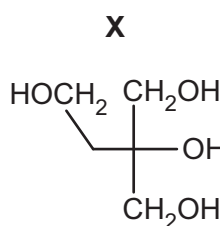
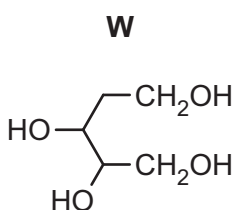
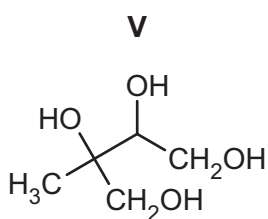
**Figure 4.1**

Context: Fig. 4.1

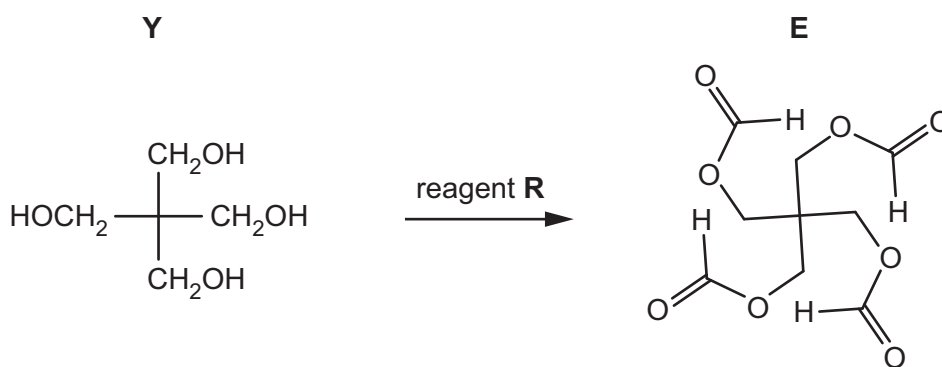
**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

- (iii)** **Y** reacts with reagent **R** to produce compound **E**.

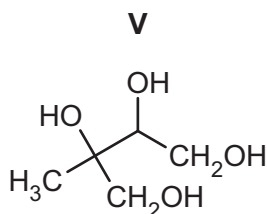
**Figure 4.3**

Name reagent **R**.

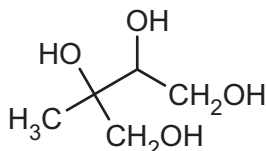
9. M/J 26/P23/Q4(d)(i)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

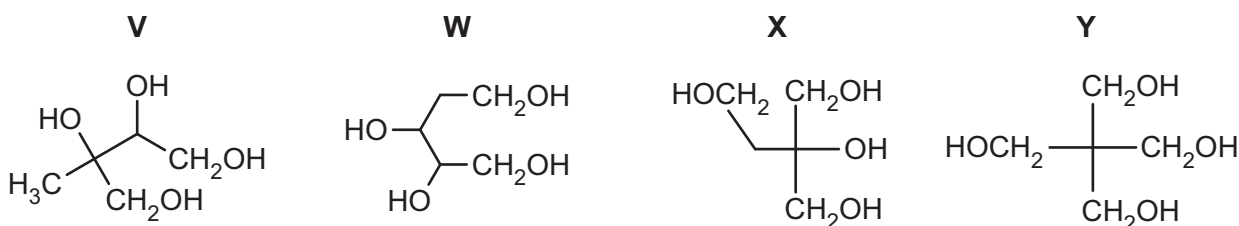
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

Context: part (d) (introduction)

- (d)** Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

- (i)** Identify which isomer, **V**, **W**, **X** or **Y**, is used in this reaction.

[1]

10. M/J 26/P24/Q4(a)

- 4** **(a)** Alcohol **S** has the structural formula $\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)\text{CH}_2\text{CH}_3$.

Classify alcohol **S**.

[1]

11. M/J 26/P24/Q4(b)

(b) Figure 4.1 shows compound **T**.

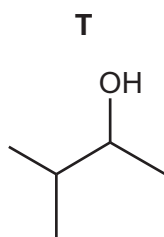


Figure 4.1

Tests are carried out on **T** using different reagents.

Complete Table 4.1 to show the observations of each test on **T**. If no reaction occurs, write 'no reaction' in the relevant box.

Table 4.1

reagent	observations
Na	
acidified $\text{K}_2\text{Cr}_2\text{O}_7$	
alkaline $\text{I}_2(\text{aq})$	

[3]

12. MAR 26/P22/Q3(e)

Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

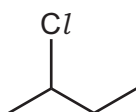


Figure 3.1

Context: part (d)

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

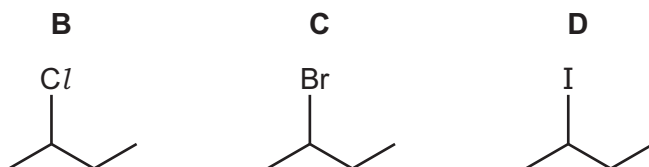


Figure 3.3

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

>

>

fastest
explanation

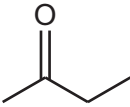
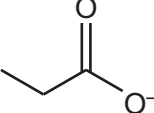
slowest

[2]

(e) The organic product of hydrolysis of each of **B**, **C** and **D** is butan-2-ol.

Complete Table 3.1 to give details of some typical chemical reactions of butan-2-ol.

Table 3.1

reagent and conditions	observation with butan-2-ol	organic product(s)
acidified $\text{K}_2\text{Cr}_2\text{O}_7$		
Na(s)	effervescence	
	yellow precipitate	 and CHI_3

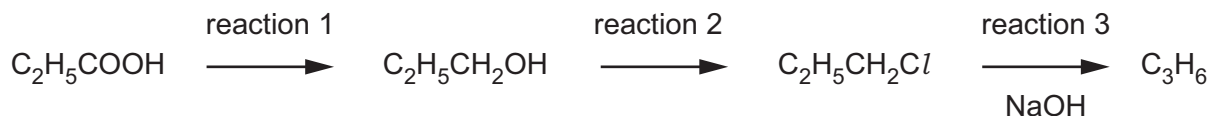
[3]

13. O/N 25/P21/Q4(a)

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

Context

- 4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

- (a) (i) Identify the type of reaction that occurs in reaction 1.

[1]

- (ii) Suggest a suitable reagent for reaction 2.

[1]

- (iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

solvent and conditions

[2]

- (iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

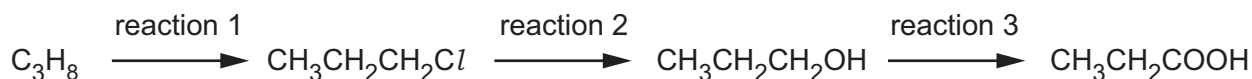
Suggest the reagent and conditions for this conversion.

[1]

14. O/N 25/P22/Q4(c)

Context

- 4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

- (c) Reaction 3 takes place when $CH_3CH_2CH_2OH$ is heated under reflux with acidified potassium dichromate(VI) solution.

- (i) State the colour change that takes place in the reaction mixture.

[1]

- (ii) Construct an equation to represent reaction 3. Use [O] to represent an atom of oxygen from the oxidising agent.

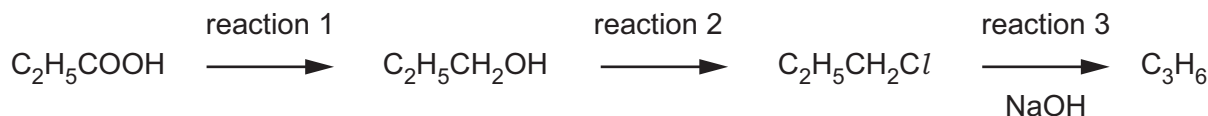
[1]

15. O/N 25/P23/Q4(a)

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

- (a) (i)** Identify the type of reaction that occurs in reaction 1.

[1]

- (ii)** Suggest a suitable reagent for reaction 2.

[1]

- (iii)** Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

solvent and conditions

[2]

- (iv)** $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

Suggest the reagent and conditions for this conversion.

[1]

16. O/N 25/P24/Q2(c)(ii)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

- (b)** A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

Context: part (c) (introduction)

- (c)** P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

- (ii)** Construct an equation for the dehydration reaction of CH_3OH to form CH_3OCH_3 .

[1]

17. O/N 25/P24/Q3(a)(i)-(iii)

Context

3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

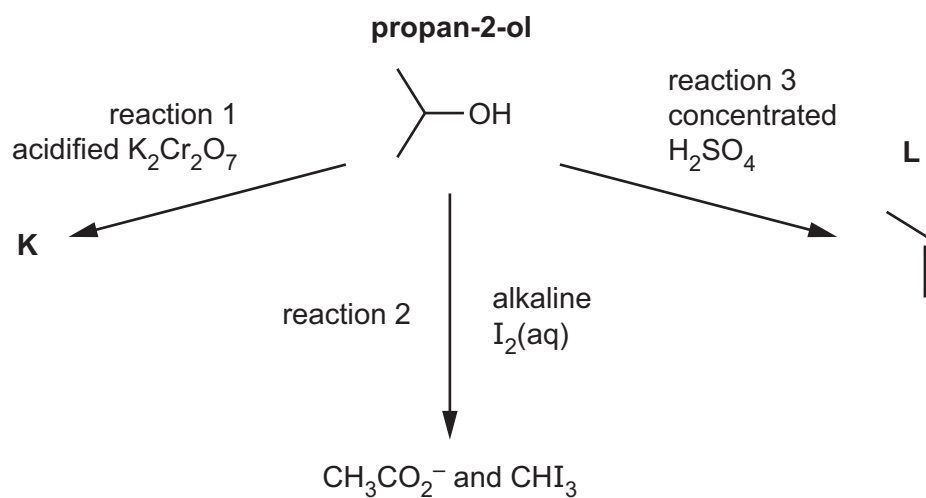


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(ii) State an observation you would make in reaction 2.

[1]

(iii) State the type of reaction that is shown in reaction 3.

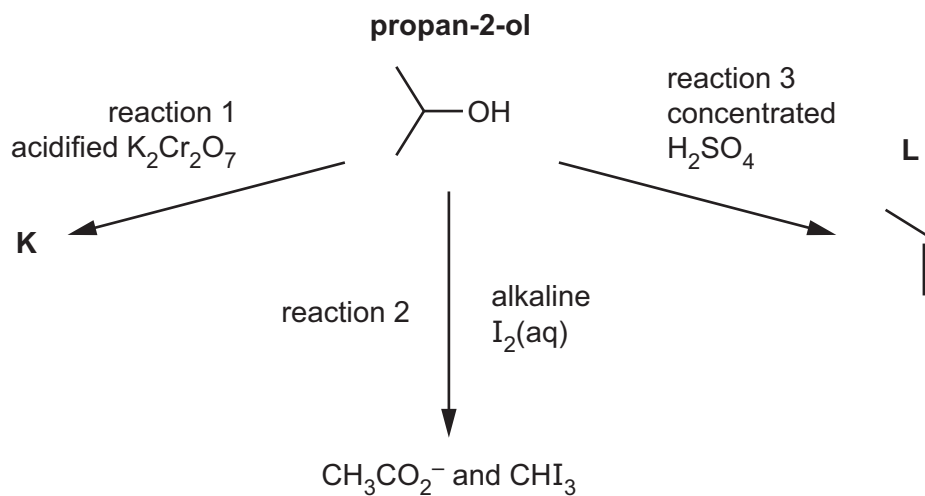
[1]

18. O/N 25/P24/Q3(b)(i)

Context

- 3** Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

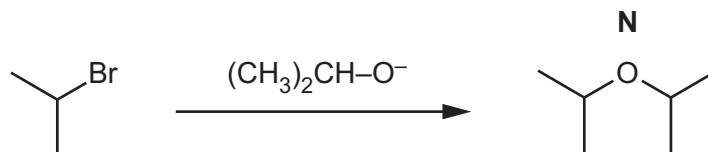
Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1**

Context: part (b) (introduction)

- (b)** Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

**Fig. 3.3**

- (i)** Write an equation for the reaction of propan-2-ol with sodium.

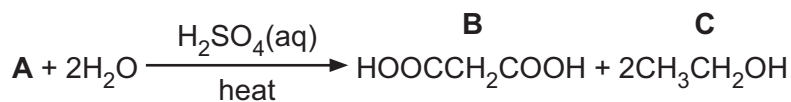
[1]

19. M/J 25/P21/Q5(c)(ii)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (c) (introduction)

- (c)** When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.

- (ii)** Suggest why **C** does **not** react with aqueous Na_2CO_3 in a similar type of reaction to **B**.

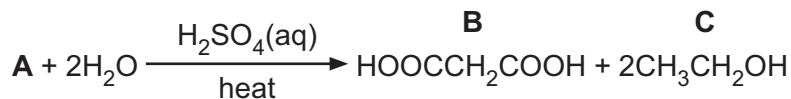
[1]

20. M/J 25/P21/Q5(d)(ii)

Context

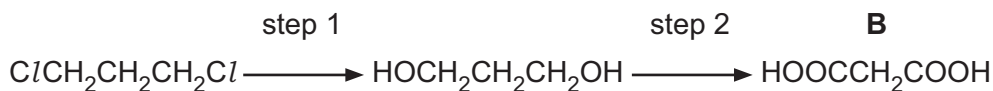
5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (d) (introduction)

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

**Fig. 5.2**

(ii) Identify the type of reaction that $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$ undergoes in step 2.

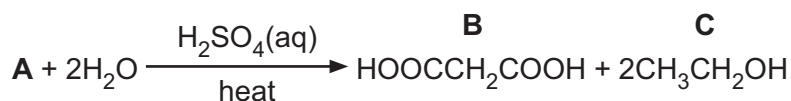
[1]

21. M/J 25/P21/Q5(e)(iii)

Context

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (e) (introduction)

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(iii) Name organic compound **E**.

[1]

22. M/J 25/P21/Q6(c)(iii)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

Context: part (c) (introduction)

- (c) Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

- (iii) Compound **J** does **not** contain the same functional group as **F**, **G** and **H**. Compound **J** also reacts with alkaline $I_2(aq)$ to produce a pale yellow precipitate.

Suggest the structure of compound **J**.

[1]

23. M/J 25/P22/Q4(b)(i)

Context: part (b) (introduction)

- (b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with $NaOH(aq)$.

- (i) Name the type of reaction that occurs in step 1.

[1]

24. M/J 25/P22/Q4(c)(ii)-(iii)

Context: part (c) (introduction)

- (c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

- (ii) Identify the gas produced when $Na(s)$ is added to separate samples of each alcohol.

[1]

- (iii) Describe the role of $Na(s)$ when it reacts with alcohols.

[1]

25. M/J 25/P22/Q4(d)

(d) Table 4.1 shows the results of two tests on separate samples of **A**, **B** and **C**.

Table 4.1

compound	heat under reflux with acidified $K_2Cr_2O_7$	warm with alkaline $I_2(aq)$
A	remains orange	no visible change
B	orange to green	no visible change
C	orange to green	pale yellow precipitate

(i) Deduce the structure of **A**.

[1]

(ii) Deduce a possible name for **B**.

[1]

(iii) Deduce the identity of **C**.

[1]

26. O/N 24/P22/Q2(b)(iv)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.

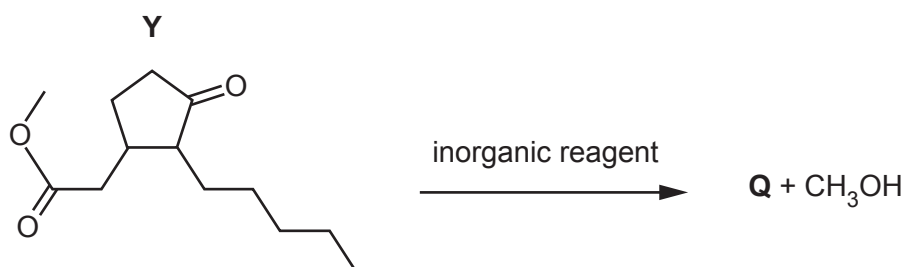
(iv) Explain why alcohols are less acidic than water.

[2]

27. M/J 24/P21/Q6(c)(iii)

Context: part (c) (introduction)

(c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

(iii) Under certain conditions, 0.0020 mol of **Q** reacts with an excess of sodium to produce a total of 44.8 cm^3 of gas at s.t.p.

Calculate the number of hydroxyl groups present in a molecule of **Q**.

Show your working.

number of hydroxyl groups =

[2]

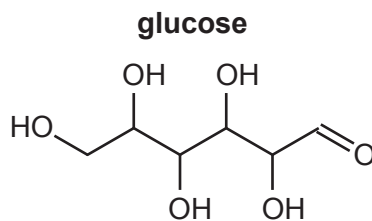
28. O/N 23/P21/Q3(e)(i)-(ii)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

- (e)** The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

[3]

29. O/N 23/P22/Q3(c)(ii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

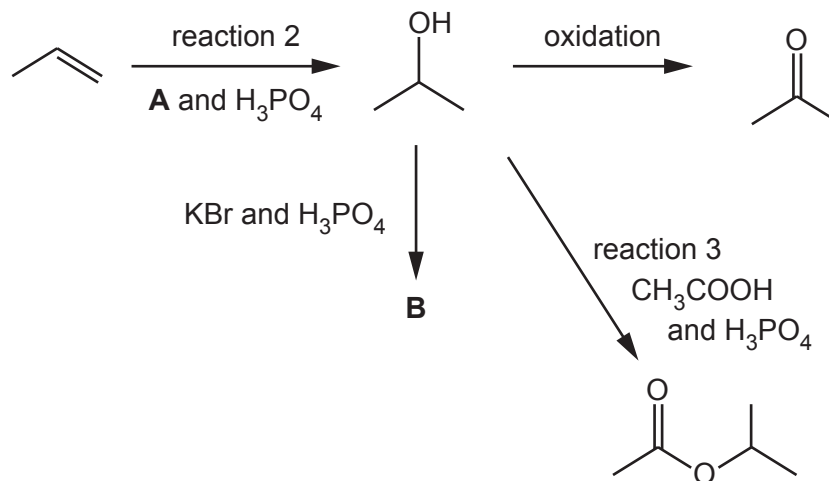


Fig. 3.1

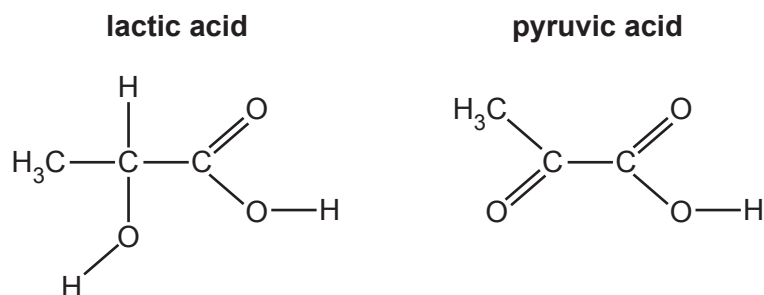
(ii) Draw the structure of **B**.

[1]

30. O/N 23/P22/Q4(b)(ii)

Context

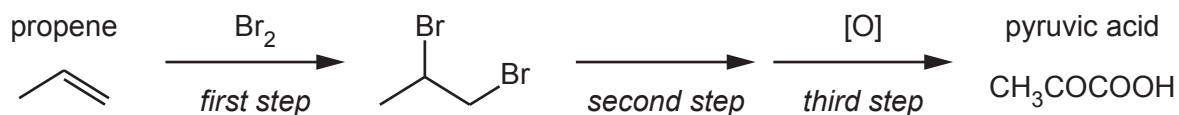
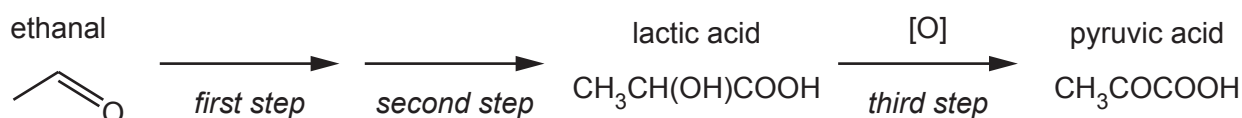
- 4** Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

**Fig. 4.1**

Context: part (b) (introduction)

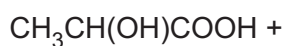
- (b)** Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (ii)** Write an equation for the oxidation of lactic acid to pyruvic acid, the third step of Fig. 4.4.

Use [O] to represent one atom of oxygen from an oxidising agent.



[1]

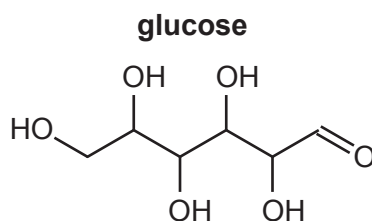
31. O/N 23/P23/Q3(e)(i)-(ii)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

- (e)** The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

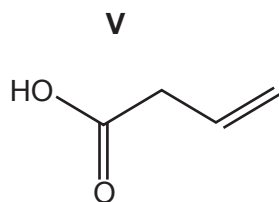
Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

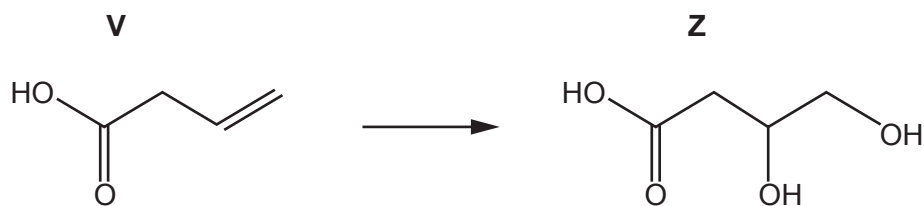
[3]

32. M/J 23/P21/Q4(b)(i)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

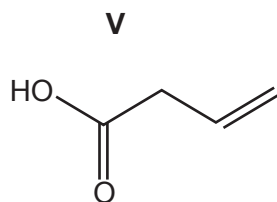
Context: part (b) (introduction)

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.**Fig. 4.2****(i)** Suggest the reagent and conditions needed to form **Z** from **V**.

[1]

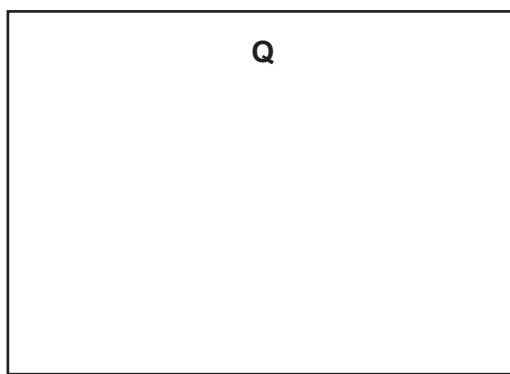
33. M/J 23/P21/Q4(c)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

- (c) **Q** contains the elements carbon, hydrogen and oxygen only. It is a saturated molecule with no branching in its carbon backbone.
Q contains only one functional group.
 The relative molecular mass of **Q** is 88.
 No effervescence is seen when Na_2CO_3 is added to **Q**.
 Effervescence is seen when sodium is added to **Q**.
Q reacts with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate.

Draw the structure of **Q** in the box.

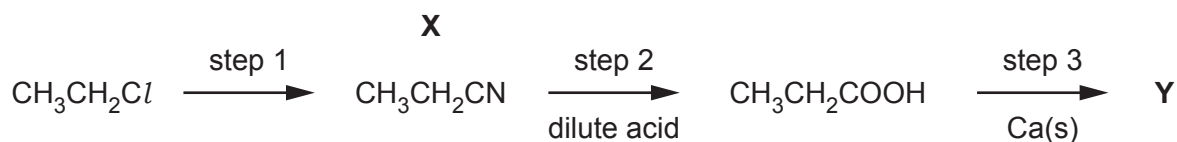


[2]

34. M/J 23/P22/Q3(b)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (b) $\text{CH}_3\text{CH}_2\text{COOH}$ can also be formed from propan-1-ol and potassium dichromate(VI).
 State the conditions required.

[1]

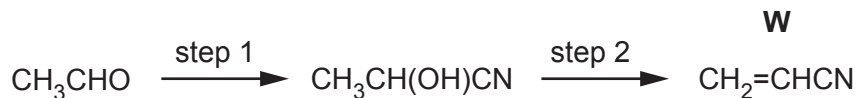
35. M/J 23/P23/Q6(b)(iii)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (iii)** Suggest a suitable reagent and conditions for step 2 in Fig. 6.1.

[1]

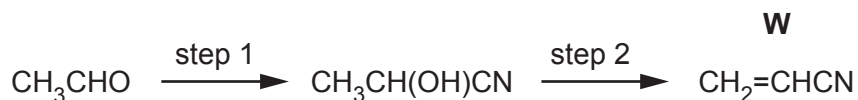
36. M/J 23/P23/Q6(f)(i)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

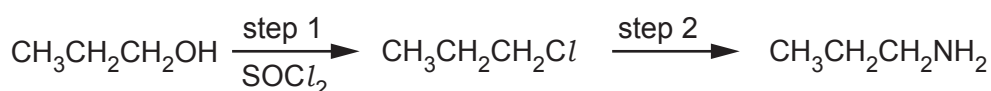
Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

Context: part (f) (introduction)

- (f)** Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

- (i)** Name the type of reaction in step 1 in Fig. 6.4.

[1]

37. MAR 23/P22/Q4(a)(iii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

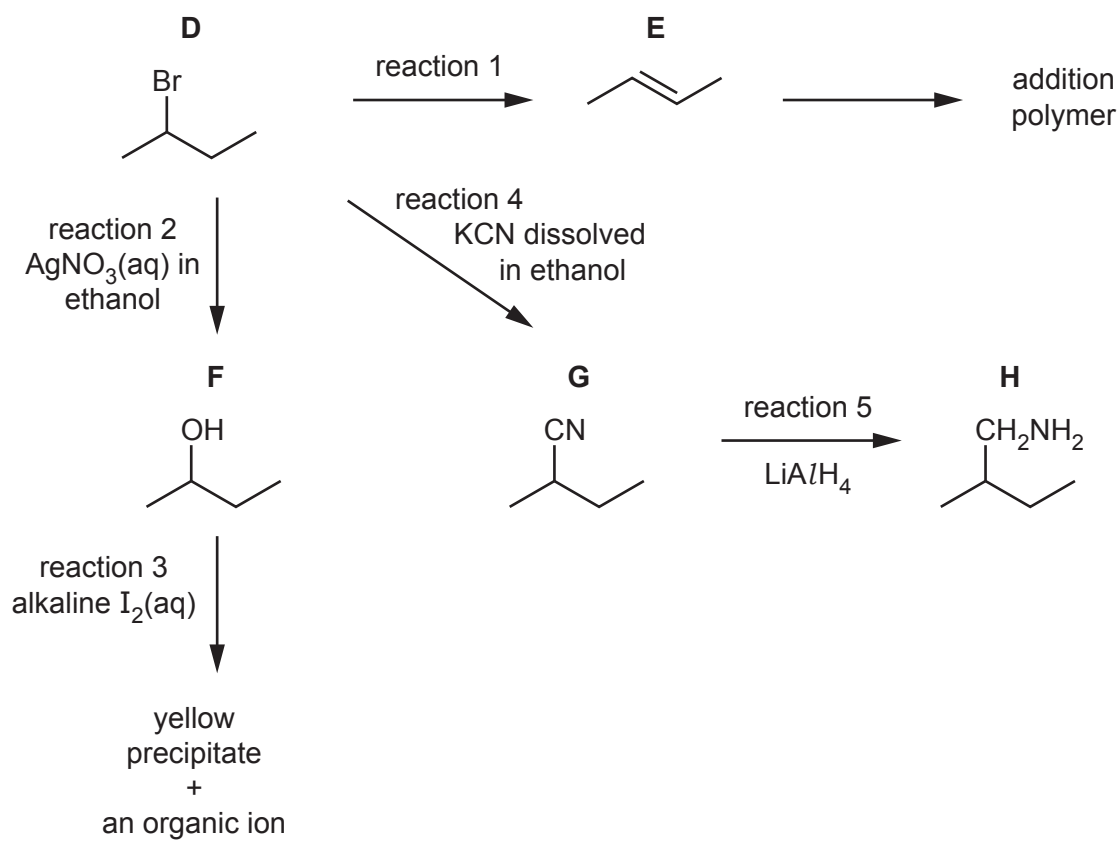


Fig. 4.1

(iii) **E** also forms when **F** is heated strongly in the presence of an Al_2O_3 catalyst.

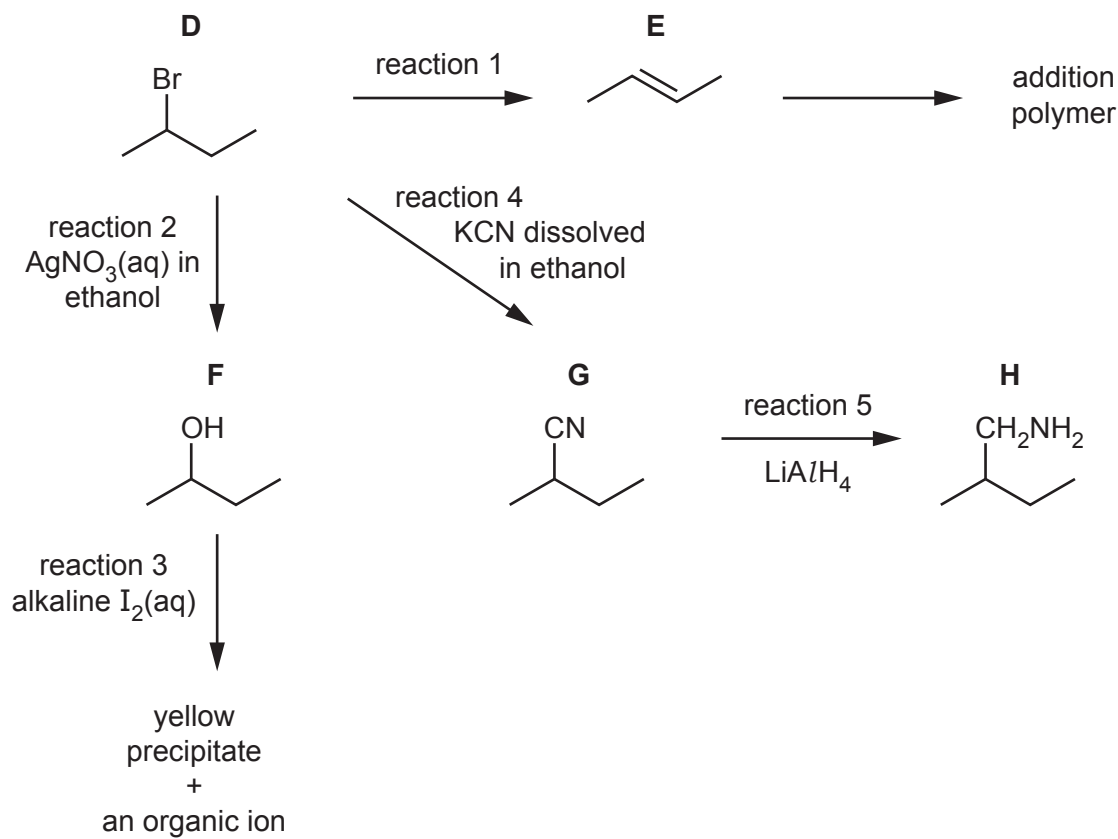
Write an equation for this reaction.

[1]

38. MAR 23/P22/Q4(b)(ii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

**Fig. 4.1**

(ii) Identify the yellow precipitate and the organic ion formed in reaction 3.

yellow precipitate

organic ion

[2]

39. O/N 22/P21/Q3(c)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(c) PCl_5 reacts with alcohols to form chloroalkanes.

(i) Identify this type of reaction.

[1]

(ii) Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

40. O/N 22/P23/Q3(c)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (c)** PCl_5 reacts with alcohols to form chloroalkanes.

- (i)** Identify this type of reaction.

[1]

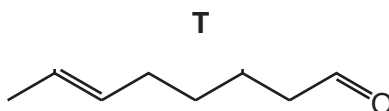
- (ii)** Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

41. M/J 22/P21/Q3(d)(i)-(ii)

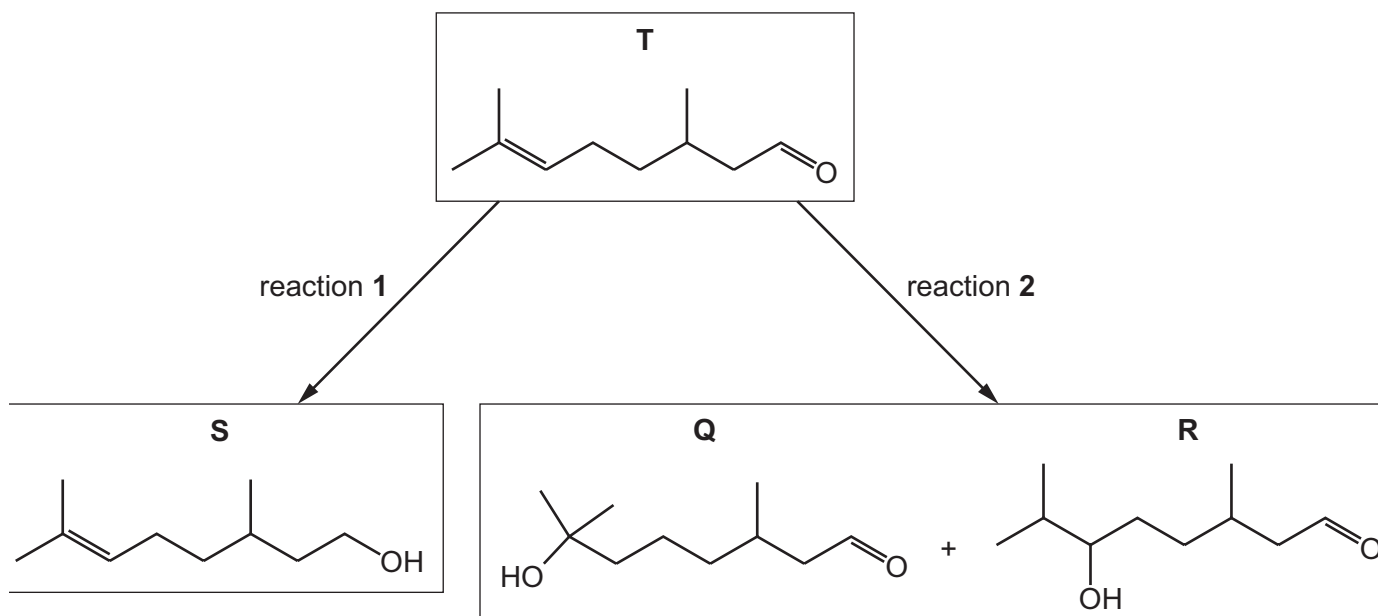
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

Context: part (d) (introduction)

- (d)** Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

Q**R**



Fig. 3.3

- (i) Predict the observations for each test-tube. Explain your answer in terms of the functional groups present in **Q** and **R**.

[3]

- (ii) When $\text{PCl}_5(\text{s})$ is added to separate samples of **Q** and **R** at room temperature, both react vigorously.

Complete the equation shown in Fig. 3.4 to describe the reaction that occurs when **R** reacts with $\text{PCl}_5(\text{s})$.

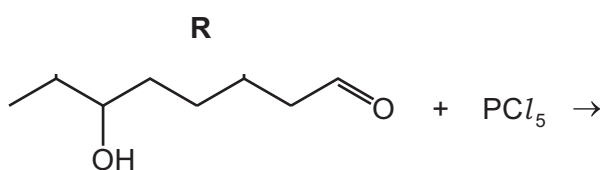


Fig. 3.4

[2]

42. M/J 22/P23/Q4(b)(iv)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula $\text{C}_4\text{H}_9\text{X}$, where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.



Fig. 4.2

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (iv) When pure **T** is added to alkaline $\text{I}_2(\text{aq})$, a yellow precipitate and an anion, **L**, are made.

Identify the anion **L**.

[1]

43. M/J 22/P23/Q5(a)(ii)

Context: part (a) (introduction)

5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.

Y does not react with Na_2CO_3 .

A gas is produced when an excess of Na is added to **Y**.

(ii) 24.0cm^3 of gas is produced when an excess of Na is added to 0.001mol of **Y**, when measured under room conditions.

Assume that 1mol of gas occupies 24.0dm^3 under room conditions.

Deduce a possible structure of **Y**. Explain your answer.

.....
[3]

Answers Section

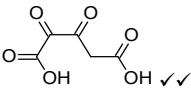
1. M/J 26/P21/Q4(a)(i)

4(a)(i)	(colour of mixture at start of reaction)orange AND (colour of mixture after the reaction)green	1 A any shade of orange A any shade of green I ppt(e) BOD blue / grey for green BOD brown for green (as dichromate is in excess)
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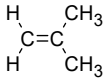
2. M/J 26/P21/Q4(b)(i)

4(b)(i)	reagent	observation with A	2 M1 I gas produced M1 I name of gas produced M1 I test for any gases M1 I solid disappears/ solid dissolves / colourless solution / flame M2 A (remains) orange solution / remains orange (solution) / no precipitate / no reaction / no observation M2 I 'turns orange'
	Na(s)	effervescence / fizzing / bubbling	
	2,4-dinitrophenylhydrazine (2,4-DNPH)	none	
	Fehling's reagent	none	

3. M/J 26/P21/Q4(b)(iii)

4(b)(iii)	$\text{HOOC}=\text{C}=\text{OCH}_2\text{COOH}$ ✓✓ OR  ✓✓ If 2 marks not awarded then max 1 for 2 of the following: • 5 carbon unbranched linear chain • 2 terminal carboxylic acid groups • 2 adjacent ketone groups only •✓	2 A any unambiguous structure I connectivity
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4. M/J 26/P22/Q4(a)

4(a)(i)	PCl_5 OR SOCl_2 OR HCl OR PCl_3 OR equivalent names	1 A thionyl chloride or sulfur oxide dichloride for SOCl_2 A phosphorus pentachloride or phosphorus trichloride BOD SOCl_2 (aq) A conc. HCl / concentrated hydrochloric acid A hydrogen chloride / HCl (g) I hydrochloric acid alone / HCl (aq) / dilute HCl I NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 R PCl_5 (aq)
4(a)(ii)	catalyst OR dehydrating agent	1 A causes dehydration A speeds up / increases rate of reaction and remains unchanged owtte A lowers activation energy / E_A I heterogenous / homogenous I acid R redox
4(a)(iii)	$\text{H}_2\text{C}=\text{C}(\text{CH}_3)_2$ OR  OR 	1 A any unambiguous structure I C_4H_8

5. M/J 26/P22/Q4(b)(i)

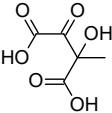
4(b)(i)	colour of mixture at start of reactionorange AND colour of mixture after the reactiongreen	1	A any shade of orange A any shade of green BOD blue / grey for green BOD brown for green (as dichromate is in excess)
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6. M/J 26/P23/Q4(a)(i)-(ii)

4(a)(i)		primary hydroxy groups	secondary hydroxy groups	tertiary hydroxy groups	2	•✓✓
	number of groups in one molecule of V	2	1	1		

4(a)(ii)			2	M1 I gas produced M1 I identity of gas M1 I test for gas M1 I solid disappears / dissolves OR colourless solution M1 I solid made M2 A (remains) orange (solution) / no precipitate / no reaction / no observation I turns orange
	reagent	observation with V		
	Na(s)	M1 effervescence / fizzing / bubbling		
	2,4-dinitrophenylhydrazine (2,4-DNPH)	M2 none		
	Fehling's reagent	none		

7. M/J 26/P23/Q4(b)

4(b)(i)	(colour of mixture at start of the reaction)orange (colour of mixture after the reaction)green	1	A any shade of orange at start A after reaction any shade of green BOD blue / grey for green BOD brown for green (dichromate is in excess)
4(b)(ii)	 OR $(\text{HOOC})\text{C}=\text{OC}(\text{COOH})(\text{OH})\text{CH}_3$	1	A any unambiguous structure I connectivity

8. M/J 26/P23/Q4(c)(iii)

4(c)(iii)	methanoic acid	1	A formic acid
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9. M/J 26/P23/Q4(d)(i)

4(d)(i)	(isomer) Y	1	A the structure of isomer Y
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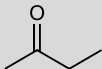
10. M/J 26/P24/Q4(a)

4(a)	tertiary / 3° (alcohol)	1	I names
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11. M/J 26/P24/Q4(b)

4(b)		3		
	reagent		observations	
	Na		M1 Bubbles / fizzing / effervescence	M1 I Na reacts / gets smaller
	acidified K ₂ Cr ₂ O ₇		M2 (Orange to) green (solution)	M2 I antiseptic smell
	alkaline I ₂ (aq)	M3 Yellow precipitate	M3 precipitate = ppt = solid = powder = crystals	

12. MAR 26/P22/Q3(e)

3(e)	reagent and conditions	observation with butan-2-ol	organic product(s)	3
	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	M1 (orange solution) turns green		
	Na(s)	effervescence	M2 	
	M3 alkaline $\text{I}_2(\text{aq})$	yellow precipitate	 and 	

Answer for context part (d)

3(d)	M1 fastest D > C > B slowest M2 carbon—halogen gets weaker down the group (so breaks faster)	2
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13. O/N 25/P21/Q4(a)

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

14. O/N 25/P22/Q4(c)

4(c)(i)	orange to green	1
4(c)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{H}_2\text{O}$	1

15. O/N 25/P23/Q4(a)

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

16. O/N 25/P24/Q2(c)(ii)

2(c)(ii)	$2\text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{OCH}_3 + \text{H}_2\text{O}$	1
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17. O/N 25/P24/Q3(a)(i)-(iii)

3(a)(i)		1
3(a)(ii)	yellow precipitate	1

3(a)(iii)	elimination	1
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18. O/N 25/P24/Q3(b)(i)

3(b)(i)	$(\text{CH}_3)_2\text{CHOH} + \text{Na} \rightarrow (\text{CH}_3)_2\text{CHO}^{\ominus}\text{Na}^{\oplus} + \frac{1}{2} \text{H}_2$	1
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19. M/J 25/P21/Q5(c)(ii)

5(c)(ii)	C is not as strong an acid as B	1
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20. M/J 25/P21/Q5(d)(ii)

5(d)(ii)	oxidation	1
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21. M/J 25/P21/Q5(e)(iii)

5(e)(iii)	1,3-dichloropropane	1
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22. M/J 25/P21/Q6(c)(iii)

6(c)(iii)	$\text{CH}_3\text{CH}(\text{OH})\text{-R}$ where R is represented as H or any alkyl group	1
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23. M/J 25/P22/Q4(b)(i)

4(b)(i)	substitution	1
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24. M/J 25/P22/Q4(c)(ii)-(iii)

4(c)(ii)	hydrogen / H_2	1
4(c)(iii)	(It / Na) reduces A / B / C / the alcohol (group) OR (it / Na behaves as) reducing agent OR (it / Na) donates electrons to A / B / C / the alcohol (group)	1

25. M/J 25/P22/Q4(d)

4(d)(i)	$(\text{CH}_3)_3\text{COH}$	1
4(d)(ii)	butan-1-ol OR (2) methylpropan-1-ol	1
4(d)(iii)	butan-2-ol OR $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	1

26. O/N 24/P22/Q2(b)(iv)

2(b)(iv)	alkyl groups are electron donating / have positive inductive effect.	1
	strengthens the O—H bond / makes H^+ less likely to be donated	1

27. M/J 24/P21/Q6(c)(iii)

6(c)(iii)	$\text{ROH} + \text{Na} \rightarrow \text{RONa} + \frac{1}{2} \text{H}_2$ (0.002 mol Q produced 0.002 mol H_2 gas so) 2 OH groups M1 answer indicates that OH group(s) in Q react with Na to produce the H_2 in the ratio 1 mol OH : $\frac{1}{2}$ mol H_2 M2 uses data to show 2OH groups	2
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28. O/N 23/P21/Q3(e)(i)-(ii)

3(e)(i)	1 4 0	1
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3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

29. O/N 23/P22/Q3(c)(ii)

3(c)(ii)		1
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30. O/N 23/P22/Q4(b)(ii)

4(b)(ii)	$\text{CH}_3\text{CH}(\text{OH})\text{COOH} + [\text{O}] \rightarrow \text{CH}_3\text{COCOOH} + \text{H}_2\text{O}$	1
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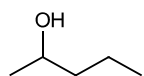
31. O/N 23/P23/Q3(e)(i)-(ii)

3(e)(i)	1 4 0	1
3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

32. M/J 23/P21/Q4(b)(i)

4(b)(i)	cold dilute acidified KMnO_4	1
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33. M/J 23/P21/Q4(c)

4(c)	 M1 molecule with 5C atoms and 1 alcohol group M2 molecule with $\text{—CH}(\text{CH}_3)\text{OH}$	2
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34. M/J 23/P22/Q3(b)

3(b)	acid (conditions) / $\text{H}^+(\text{aq})$ AND heat under reflux	1
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35. M/J 23/P23/Q6(b)(iii)

6(b)(iii)	heat + conc. sulfuric acid / conc. phosphoric acid OR heat + Al_2O_3	1
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36. M/J 23/P23/Q6(f)(i)

6(f)(i)	substitution	1
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37. MAR 23/P22/Q4(a)(iii)

4(a)(iii)	$\text{C}_4\text{H}_{10}\text{O} / \text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5 \rightarrow \text{C}_4\text{H}_8 / \text{CH}_3\text{CHCHCH}_3 + \text{H}_2\text{O}$	1
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38. MAR 23/P22/Q4(b)(ii)

4(b)(ii)	CHI_3 / iodoform / tr(i)iodomethane	1
	$\text{CH}_3\text{CH}_2\text{CO}_2^-$ / propanoate	1

39. O/N 22/P21/Q3(c)

3(c)(i)	substitution	1
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3(c)(ii)		1
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40. O/N 22/P23/Q3(c)

3(c)(i)	substitution	1
3(c)(ii)		1

41. M/J 22/P21/Q3(d)(i)-(ii)

3(d)(i)	M1 Q orange → green M2 R orange → green M3 –CHO / aldehyde (in both) (and 2° / secondary alcohol in R) reacts / oxidised	3
3(d)(ii)	 M1 correct formula of organic product M2 correct inorganic products	2

42. M/J 22/P23/Q4(b)(iv)

4(b)(iv)	$\text{C}_2\text{H}_5\text{COO}^-$ OR $\text{CH}_3\text{CH}_2\text{CO}_2^-$ or propanoate (an)ion	1
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43. M/J 22/P23/Q5(a)(ii)

5(a)(ii)	M1 unbranched 4C structure AND any number of –OH in any position M2 0.001 mol H_2 made from 0.001 mol Y AND $\text{R-OH} + \text{Na} \rightarrow \text{RONa} + \frac{1}{2}\text{H}_2$ OR use of 1 OH (group) $\rightarrow \frac{1}{2}\text{H}_2$ M3 $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$	3
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